

Context-Aware Predictive Customer Churn Engine

Comprehensive Problem Statement & Project Overview

Python XGBoost MCP Framework FastAPI Docker Machine Learning

1. Executive Summary

Customer retention is fundamentally more cost-effective than customer acquisition, yet businesses consistently struggle to identify at-risk users before they abandon a platform. The **Context-Aware Predictive Customer Churn Engine** closes the gap between raw predictive analytics and interactive customer success workflows. The project developed an end-to-end machine learning pipeline — built through exploratory data analysis, feature engineering, and hyperparameter tuning — that predicts customer churn with an **85% F1-score** while identifying the key retention drivers behind each prediction. That model was then engineered into an AI system that exposes churn prediction as an **MCP (Model Context Protocol) tool**, enabling Large Language Models to invoke live inference dynamically during a conversation. This transforms a static ML model into a conversational, context-aware AI agent rather than a passive analytics report.

2. The Problem Statement

The Core Business Challenge

In subscription-based and SaaS business models, customer churn directly degrades Annual Recurring Revenue (ARR). Historical data contains the signals necessary to predict when a user is likely to churn — declining usage metrics, increased support tickets, payment failures — but identifying these signals manually is impossible at scale.

The Technical Disconnect

Traditional machine learning solutions approach this problem by generating static churn probability reports. These deployments suffer from three critical flaws:

Siloed Intelligence — Churn models typically live in isolated data warehouses or BI dashboards, disconnected from the live operational tools used by customer success teams or conversational AI agents.

Lack of Real-Time Context — When an AI agent or a support representative interacts with a user, they have no on-demand access to that user's predictive churn risk profile mid-conversation.

Inactionable Outputs — Knowing a customer might churn is insufficient on its own; agents need to know *why* — the key retention drivers — to take corrective action instantly.

Objective: How can a highly accurate churn prediction model be democratized by embedding it directly into the context window of conversational LLMs — via an MCP tool exposing live inference — to enable proactive, automated customer retention strategies?

3. Proposed Solution Architecture

Phase 1 — Advanced Predictive Modeling

The foundation of the system is a rigorously engineered ML pipeline. Exploratory Data Analysis (EDA) and feature engineering transformed raw behavioral and transactional data into predictive signals. An XGBoost classifier was selected for its performance on tabular data and its native feature-importance output. Extensive hyperparameter tuning optimized the model to maximize recall on the minority "churn" class while preserving precision, ultimately achieving an **85% F1-score** and surfacing the key retention drivers behind each prediction.

Phase 2 — High-Performance Serving Layer

The trained pipeline is wrapped in an asynchronous REST API built with **FastAPI**. The API accepts customer data payloads, runs them through the pre-trained XGBoost model, and returns churn probabilities alongside top contributing factors. The application is containerized with **Docker**, ensuring environment parity across development, testing, and production and enabling scalable deployment.

Phase 3 — MCP Integration: Exposing the Model as an AI Tool

The core engineering contribution of this project is exposing the FastAPI churn-prediction endpoint as an **MCP (Model Context Protocol) tool**. Rather than requiring users to query an API manually, MCP lets an LLM natively understand and invoke the endpoint's capabilities. During a live conversation, the LLM autonomously calls the churn prediction tool, retrieves the live inference result along with its retention drivers, and uses that context to alter its own conversational strategy — for example, offering a targeted discount or escalating to a human manager.

4. Key Deliverables & Impact

85% F1-Score Achieved · Live LLM Inference Capability

End-to-End ML Pipeline — Full pipeline from EDA, feature engineering, and hyperparameter tuning through to a production-ready, containerized model, achieving an 85% F1-score.

Actionable Retention Insights — The model identifies and highlights key retention drivers, not just churn probability, giving the business root-cause visibility.

AI System with Live MCP Inference — An AI system exposing the churn model as an MCP tool, enabling LLMs to invoke live inference dynamically and act on it mid-conversation — turning a reactive dashboard into a proactive, conversational retention engine.